Juan José Duque

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PROFILE

Backend-focused Full Stack Developer with 4+ years of experience building scalable, production-grade systems. Specialized in Python and Node.js for API development, microservices, and asynchronous processing pipelines. Proven track record improving system performance (40–50% latency reduction) through query optimization, data modeling, and efficient backend architecture. Strong experience working with real-time data systems, integrations, and high-volume applications. Fluent in English (C1), comfortable working in international and remote environments.

WORK EXPERIENCE

Full Stack Developer (Backend-focused) *Remote*
Unitti *2021 - 2026*

- Designed and developed Python-based microservices for real-time data processing, improving system scalability and reliability in production.
- Built and maintained REST APIs and backend services using Python and Node.js, handling business logic, integrations, and data workflows.
- Optimized PostgreSQL queries and system bottlenecks, reducing latency by 20–40% and improving overall throughput.
- Implemented asynchronous processing pipelines to handle heavy workloads without blocking user requests.
- Diagnosed and resolved complex production issues related to real-time data flows, improving system stability and reliability.

EDUCATION

Universidad del Valle, Cali, Colombia
Information Systems Engineering

TECHNICAL SKILLS

Backend: Python, Node.js, REST APIs, Microservices, Async Processing, Data Pipelines
Databases: PostgreSQL, MongoDB, SQL Server, Query Optimization, Data Modeling
Cloud & DevOps: AWS (S3, SQS), Docker, CI/CD, Linux, Git
Frontend: Vue 3, React, TypeScript (for integration and full-stack capabilities)
Testing: Pytest, Jest, Unit & Integration Testing, Postman
Tools: Jira, Figma, Datadog

PROJECTS

Asynchronous Report Processing System Technologies: Python, PostgreSQL

- Designed and implemented an asynchronous processing system using background workers and task queues.
- Reduced report generation time from minutes to seconds by decoupling heavy workloads from request lifecycle.
- Improved reliability with retry mechanisms, error handling, and job tracking.

Reusable Google Maps Integration Architecture Technologies: JavaScript, APIs, Microservices, AWS

- Designed a reusable and scalable integration architecture used across multiple products.
- Built modular services for handling dynamic data visualization and real-time updates.
- Reduced development time by standardizing map-based feature implementation across teams.

LANGUAGES

Spanish (Native), English (C1 – Professional Working Proficiency)